ENTROPY-GATED TRIPLET CROSS-MEMORY-BANK LEARNING FOR LONG-TAILED MULTI-LABEL CHEST X-RAY CLASSIFICATION

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ABSTRACT

ConvNeXt with a query-based decoder is a simple, efficient chest X-ray classifier. Our previous CXR-LT 2026 entry improved it with a sparse mixture of experts (MoE) that adds parameters at inference and was reported with aggregate metrics only, so its effect on rare findings was not shown. We study training-time alternatives that leave the backbone unchanged: a triplet loss against a cross-batch memory bank with anchors chosen by predictive entropy, and a small patient-history Markov prior. Under one recipe, five seeds and seven metrics we compare the MoE, the entropy-gated triplet memory and the same with the Markov prior. **[TBD: results]**

***Index Terms*—** Chest X-ray, long-tailed classification, triplet loss, cross-batch memory, predictive entropy

1. INTRODUCTION

Findings on chest radiographs are long-tailed and co-occur: in the CXR-LT 2026 Task 1 data [1], built on PadChest [2], three of the 30 findings occur in over 10% of our training images and seven in under 1%. A ConvNeXt [3] with a query-based ML-Decoder head [4] trained with Asymmetric Loss (ASL) [5] is a simple, efficient alternative to large transformers [6].

Our previous entry [7] improved this model with a sparse MoE in the top stages and the decoder (third place, 0.4599 test mAP). Two problems remain: the experts and router add parameters at inference, and only aggregate metrics were reported, without a matched baseline or head, medium and tail results, so it is unknown whether the MoE helps rare findings or beats the plain model.

Here we ask whether the same ConvNeXt can instead be improved without extra inference-time parameters, by giving rare, co-occurring findings a stronger training signal and by using a patient's earlier studies. Therefore, in this paper we introduce new techniques to better fine-tune and improve ConvNeXt, a simple yet efficient model.

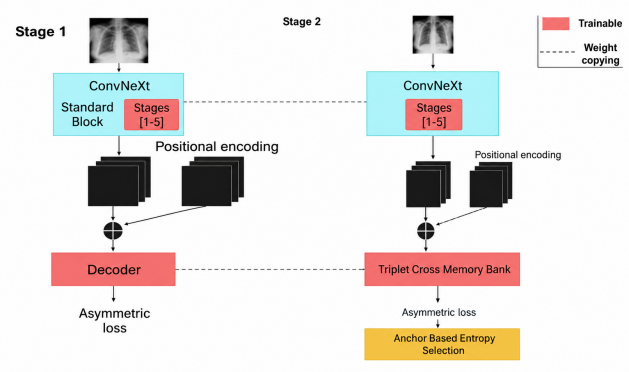


Fig. 1. Two-stage training: Stage 1 with ASL only; Stage 2 adds the entropy-gated triplet loss with a cross-batch memory bank (Sec. 2).

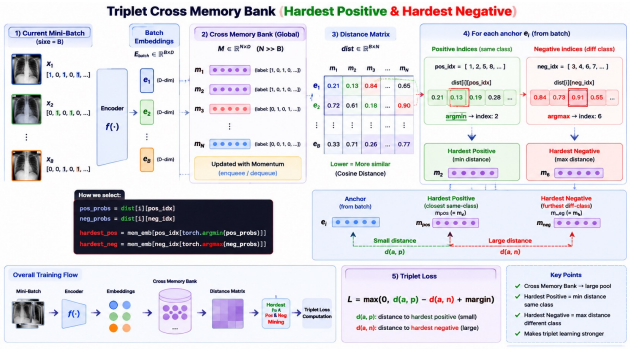


Fig. 2. Triplet cross-memory-bank pipeline (Eqs. (3)–(4)). In the drawing, the "hardest" pairs are the easiest in the usual terminology (Sec. 2.4), and the "momentum" update is the plain first-in-first-out queue of Sec. 2.2.

2. METHOD

2.1. Model

The classifier is a ConvNeXt-B [3] with an added fifth stage of two blocks (total stride 64), followed by an ML-Decoder

head [4]: the final feature map receives a 2D positional encoding [8] and is projected to 768 channels, 30 fixed (non-trainable) label queries pass through two decoder layers, each with cross-attention to the feature map followed by self-attention among the queries, and a group-wise linear layer gives one logit z_{ic} per finding (Fig. 1). Stage 1 trains this model end-to-end with ASL at 384 px [7]; Stage 2 fine-tunes all of its trainable weights from that checkpoint. Only Stage 2 differs from our earlier entry. For metric learning we average the 30 decoded query embeddings of image i and ℓ_2-normalise the result, giving $e_i \in \mathbb{R}^{768}$.

2.2. Cross-batch memory bank

With batches of eight, few positive and negative pairs are available inside a batch, and the hardest pair found in a batch approximates the hardest pair in the data poorly. A first-in-first-out bank $\mathcal{M}$ holds the embeddings and multi-hot label vectors $\mathbf{y}$ of the 2,048 most recent training images (256 mini-batches of eight). After every step the batch's embeddings, detached from the graph, and labels replace the oldest entries, so each anchor sees a pool 256 times the size of a batch at no extra forward cost [9]. The entries come from slightly older encoder states.

2.3. Entropy-gated anchors

Rather than mining a triplet for every sample in the batch, we select anchors using the model's own predictive uncertainty, the criterion of active learning [10]. For each sample x_i in a mini-batch $\mathcal{B}$ we compute the Shannon entropy of the softmax distribution over its $C = 30$ classification logits, $p_\theta(y_c \mid x_i) = \exp z_{ic} / \sum_k \exp z_{ik}$,

$$H(x_i) = -\sum_{c=1}^{C} p_\theta(y_c \mid x_i)\, \log p_\theta(y_c \mid x_i), \tag{1}$$

and select as anchors the samples whose entropy is at or above the batch mean,

$$\mathcal{A} = \Big\{ x_i \in \mathcal{B} : H(x_i) \;\geq\; \tfrac{1}{|\mathcal{B}|} \textstyle\sum_{x_j \in \mathcal{B}} H(x_j) \Big\}. \tag{2}$$

On average 4.2 of the eight images in a batch pass the gate (4.7 in the first epoch, 3.4 in the last epoch trained; two reruns of arm C), and 55% of these anchors find a valid positive and negative in the bank. The rationale is that an image classified with low entropy is already resolved and less informative to move in embedding space, so the triplet term should go to uncertain images; the ablation tests this, we do not assume it. Eq. (1) treats the findings as one categorical distribution although they are separate binary labels; it is the rule we ablate, and a per-label (Bernoulli) entropy is a natural alternative.

2.4. Mining and objective

For an anchor $x_i \in \mathcal{A}$ with embedding e_i and multi-hot label vector $\mathbf{y}_i$, positives are bank entries that share at least one finding with x_i other than the five most frequent ones (Normal, pleural effusion, cardiomegaly, support devices, aortic elongation), and negatives share no finding (Fig. 2):

$$\begin{aligned} \mathcal{P}(i) &= \{ m \in \mathcal{M} : \mathbf{y}_m^\top D\, \mathbf{y}_i \geq 1 \}, \\ \mathcal{N}(i) &= \{ m \in \mathcal{M} : \mathbf{y}_m^\top \mathbf{y}_i = 0 \}, \end{aligned} \tag{3}$$

where the diagonal matrix D is zero at the five most frequent findings, so that two studies that are both Normal are not a positive pair. We take the closest positive and the farthest negative, $e_i^+ = \arg\min_{m \in \mathcal{P}(i)} \|e_i - e_m\|_2$ and $e_i^- = \arg\max_{m \in \mathcal{N}(i)} \|e_i - e_m\|_2$. These are easy pairs rather than the hardest ones [11]; they are less exposed to noise in labels derived from reports, but we did not compare harder mining. Over the anchors $\mathcal{A}' \subseteq \mathcal{A}$ for which both sets are non-empty,

$$\mathcal{L}_{\text{tri}} = \frac{1}{|\mathcal{A}'|} \sum_{x_i \in \mathcal{A}'} \max\big(0, \|e_i - e_i^+\|_2 - \|e_i - e_i^-\|_2 + \alpha\big), \tag{4}$$

and the objective is $\mathcal{L} = \mathcal{L}_{\text{ASL}} + \lambda\, \mathcal{L}_{\text{tri}}$ with $\alpha = 0.5$ and $\lambda = 0.1$. $\mathcal{L}_{\text{ASL}}$ uses $\gamma_- = 2$, $\gamma_+ = 0$, a probability margin of 0.05 and positive-term class weights proportional to $\log(1/f_c)$, with f_c the training prevalence, normalised to mean one and clipped to $[0.5, 2]$, so that rare findings contribute a larger positive gradient without destabilising the frequent ones. The bank, gate and triplet term exist only during training.

2.5. Patient-history Markov prior

Findings persist across a patient's studies. For each finding c, a two-state Markov chain estimated on the training fold gives its probability $\pi_c(y_{t-1})$ at a study from its presence at the patient's previous study. With $\bar{\pi}_c$ the training prevalence, the prior vector is $r_c = \operatorname{logit} \pi_c(y_{t-1}) - \operatorname{logit} \bar{\pi}_c$, with a flag for an earlier study appended and $r = 0$ when there is none. It offsets the label queries Q_c before they attend to the image (Fig. 3):

$$Q_c \leftarrow Q_c + r_c\, u_c + W r, \qquad c = 1, \ldots, 30, \tag{5}$$

where u_c and W are zero-initialised, so training starts from the model without the prior. The objective, the entropy gate and the memory bank are unchanged: the offset acts before cross-attention, so it shifts the logits whose entropy selects the anchors and the decoded query embeddings that fill the bank. A post-prediction patient HMM that adds patient metadata uses the same chain (Fig. 4). Results are in Table 1.

3. RESULTS

Our earlier entry, a sparse mixture of experts [7], reported 0.405 mAP on this internal split at 384 px and 0.4599 on the

Table 1. Unified test on the internal validation split (17,270 images): 768 px, five seeds, mean ± s.d. All systems share the Stage-1 checkpoint, the schedule (four of eight cosine epochs, drop-path 0.1), SWA weights and flip test-time augmentation. Markov prior: history available, or withheld as on the challenge test set. Head, medium, tail: mean AP over 3, 20, 7 findings. mECE: lower is better. Empty cells await the runs.

System	n	mAP	mAUC	mF1	mECE	head AP	medium AP	tail AP
MoE (sparse mixture of experts)								
Triplet Memory + Entropy								
Triplet Memory + Entropy + Markov prior, history available								
Triplet Memory + Entropy + Markov prior, history withheld								

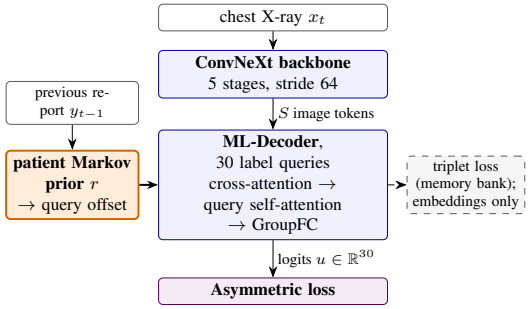


Fig. 3. Where the Markov prior enters (orange): the patient-history prior r offsets the ML-Decoder label queries.

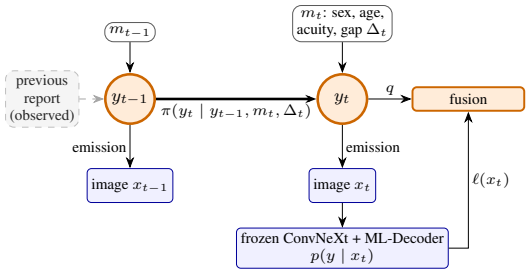


Fig. 4. Post-prediction patient HMM: hidden finding states y_t, images and the frozen classifier as emissions, metadata and time gap Δ_t in the transition prior.

official test set (third place); that recipe differs from ours, so these numbers are not comparable with Table 1. **[TBD: results text after the unified test: the three systems, paired differences over seeds with 95% intervals and exact sign-flip tests, head, medium and tail AP, calibration]**

4. DISCUSSION AND LIMITATIONS

[TBD: interpretation of the unified test] An earlier six-seed ablation at 512 px, run before this test, found no measurable effect of the entropy gate over using every image as an anchor. All results come from one internal validation fold that shares patients with the training set (77% of its images) and includes lateral views (30%), whereas the challenge test set is frontal and patient-disjoint; the absolute values are therefore likely optimistic. The Markov prior needs an earlier report of the same patient, which the test set does not provide, so only its withheld rows are test-like. The entropy in Eq. (1) is a softmax over separate binary labels.

5. CONCLUSION

[TBD: conclusion after the unified test]

6. COMPLIANCE WITH ETHICAL STANDARDS

This research study was conducted retrospectively using human subject data made available in open access by the CXR-LT 2026 challenge [1], which is built on PadChest [2]. Ethical approval was not required as confirmed by the license attached with the open access data.

7. ACKNOWLEDGMENTS

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